

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Fundamentals of Data Science COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO3: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 01 Total Marks:100 Annexure A

CASE STUDY

Code of Conduct:

I Anto Chriswin / 192324093 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Levels	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly		
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts		
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning		
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided		

Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand		
---------	----	--	---	-------------------------------	---------------------------------------	--	--

Annexure A

Case Study 1: E-Commerce Customer Analysis

Scenario:

An e-commerce company has collected customer transaction data, including purchase amount, number of orders, browsing time, and customer ratings. The company wants to understand customer purchasing behavior and improve its marketing strategies.

Questions:

- How can descriptive statistics such as mean, median, standard deviation, and quartiles be used to summarize customer transaction data?
- Discuss how unusually high or low purchase amounts can affect the analysis and describe suitable methods for detecting outliers.
- How can hypothesis testing be used to determine whether premium members spend significantly more than regular customers? Formulate the hypotheses and explain the analysis process.

Answer :

A. Descriptive Statistics

Descriptive statistics help summarize and understand customer transaction data. The **mean** shows the average purchase amount, number of orders, browsing time, and ratings. The **median** represents the middle value and is useful when the data contains extreme values. **Standard deviation** measures how much customer spending varies from the average, while **quartiles** divide the data into four parts, helping identify customer segments and detect unusual purchase patterns. Together, these measures provide a clear overview of customer purchasing behavior.

B. Effect of Outliers and Detection Methods

Unusually high or low purchase amounts (outliers) can distort the analysis by affecting the mean and increasing the standard deviation. This may lead to incorrect conclusions about customer spending habits and marketing strategies. Outliers can be detected using the **Interquartile Range (IQR) method**, where values outside $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$ are considered outliers, or the **Z-score method**, where values with $|Z| > 3$ are treated as outliers. Detecting outliers improves the accuracy of data analysis.

C. Hypothesis Testing

Hypothesis testing is used to determine whether premium members spend significantly more

than regular customers. The hypotheses are:

- **Null Hypothesis (H_0):** Premium members do not spend significantly more than regular customers.
- **Alternative Hypothesis (H_1):** Premium members spend significantly more than regular customers.

An **Independent Sample t-test** is performed using the purchase data of both groups. The **p-value** is compared with the significance level ($\alpha = 0.05$). If $p < 0.05$, the null hypothesis is rejected, indicating that premium members spend significantly more. Otherwise, there is insufficient evidence to conclude a significant difference.

Case Study 2: Renewable Energy Production Analysis

Scenario:

A solar power company has collected data from multiple solar plants, including daily energy generation, sunlight hours, panel temperature, and system efficiency. The company wants to evaluate plant performance and verify whether the average energy production meets the expected target.

Questions:

- Explain how descriptive statistics and distribution analysis can be used to summarize and evaluate the energy production dataset.
- Discuss the impact of abnormal energy generation values on the analysis and describe methods for detecting and handling outliers.
- How can confidence intervals and hypothesis testing be used to determine whether the average daily energy generation differs significantly from the expected target value?

Answer :

A. Descriptive Statistics and Distribution Analysis

Descriptive statistics help summarize the energy production data from different solar plants.

The **mean** shows the average daily energy generation, **median** gives the middle value, **standard deviation** measures the variation in energy production, and **quartiles** divide the data into four parts to understand its spread. **Distribution analysis** helps identify whether the data is normally distributed, skewed, or contains unusual patterns, enabling better evaluation of plant performance.

B. Impact of Outliers and Detection Methods

Abnormal energy generation values may occur due to equipment failure, poor weather conditions, or sensor errors. These outliers can distort the average energy production and reduce the accuracy of performance analysis. Outliers can be detected using the **Interquartile Range (IQR) method** or the **Z-score method**. After detection, they can be verified, corrected

if caused by errors, or removed if they negatively affect the analysis.

C. Confidence Intervals and Hypothesis Testing

A **confidence interval** estimates the range within which the true average daily energy generation is likely to lie. If the expected target falls outside this interval, it may indicate a significant difference. Hypothesis testing is used to verify this statistically.

- **Null Hypothesis (H_0):** The average daily energy generation is equal to the expected target.
- **Alternative Hypothesis (H_1):** The average daily energy generation is different from the expected target.

A **one-sample t-test** is performed, and the **p-value** is compared with **0.05**. If $p < 0.05$, reject H_0 and conclude that the average energy generation differs significantly from the expected target; otherwise, fail to reject H_0 .

Case Study 3: Hospital Patient Recovery Analysis

Scenario:

A hospital has collected patient recovery data, including age, treatment cost, number of hospital days, recovery time, and patient satisfaction score. The hospital wants to analyze treatment effectiveness and improve patient care quality.

Questions:

- How can descriptive statistics such as mean, median, standard deviation, and quartiles be used to summarize patient recovery time and treatment cost?
- Discuss how unusually high hospital stay duration or treatment cost values may affect the analysis. What methods can be used to detect outliers?
- How can hypothesis testing be used to determine whether patients receiving a new treatment recover faster than patients receiving the existing treatment? Formulate the hypotheses and explain the analysis process.

Answer :

A. Descriptive Statistics

Descriptive statistics help summarize patient recovery data. The **mean** shows the average recovery time and treatment cost, the **median** gives the middle value and is less affected by extreme values, the **standard deviation** measures the variation in recovery time and costs, and **quartiles** divide the data into four parts to identify the spread and detect unusual recovery times or treatment costs. These measures help evaluate treatment effectiveness and patient care quality.

B. Impact of Outliers and Detection Methods

Unusually long hospital stays or very high treatment costs can distort the average and increase data variability, leading to inaccurate conclusions about patient recovery. Outliers can be detected using the **Interquartile Range (IQR) method** or the **Z-score method**. Once identified, they should be verified and either corrected, analyzed separately, or removed if they result from data errors.

C. Hypothesis Testing

Hypothesis testing is used to determine whether patients receiving the new treatment recover faster than those receiving the existing treatment.

- **Null Hypothesis (H_0):** There is no significant difference in recovery time between the new and existing treatments.
- **Alternative Hypothesis (H_1):** Patients receiving the new treatment recover significantly faster than those receiving the existing treatment.

An **Independent Sample t-test** is performed by comparing the recovery times of the two groups. The **p-value** is compared with the significance level ($\alpha = 0.05$). If $p < 0.05$, reject H_0 and conclude that the new treatment results in faster recovery. Otherwise, fail to reject H_0 .

Case Study 4: Student Academic Performance Analysis

Scenario:

A university has collected student performance data, including attendance percentage, assignment marks, internal test scores, study hours, and final exam marks. The university wants to identify factors affecting student success.

Questions:

- Explain how descriptive statistics and distribution analysis can be used to summarize student marks and attendance data.
- Discuss how extremely low or high marks can affect the interpretation of academic performance. Describe suitable methods for detecting and handling outliers.
- How can hypothesis testing be used to check whether students with attendance above 75% score significantly higher than students with attendance below 75%? Formulate the hypotheses and explain the steps.

Answer:

A. Descriptive Statistics and Distribution Analysis

Descriptive statistics help summarize student performance data. The **mean** shows the average marks and attendance, the **median** gives the middle value, **standard deviation** measures the variation in marks, and **quartiles** divide the data into four parts to understand its spread.

Distribution analysis helps identify whether marks are normally distributed, skewed, or contain unusual values, making it easier to evaluate student performance.

B. Impact of Outliers and Detection Methods

Extremely high or low marks can distort the average and affect the interpretation of student performance. They may also increase the variation in the data and lead to misleading conclusions. Outliers can be detected using the **Interquartile Range (IQR) method** or the **Z-score method**. After detection, they should be verified and handled by correcting errors, removing invalid values, or analyzing them separately.

C. Hypothesis Testing

Hypothesis testing is used to determine whether students with attendance above **75%** score significantly higher than those with attendance below **75%**.

- **Null Hypothesis (H_0):** There is no significant difference in exam scores between students with attendance above 75% and below 75%.
- **Alternative Hypothesis (H_1):** Students with attendance above 75% score significantly higher than those with attendance below 75%.

An **Independent Sample t-test** is performed by comparing the exam scores of the two groups. The **p-value** is compared with **0.05**. If **$p < 0.05$** , reject H_0 and conclude that higher attendance leads to significantly higher scores. Otherwise, fail to reject H_0 .

Case Study 5: Banking Loan Approval Analysis

Scenario:

A bank has collected customer loan application data, including income, credit score, loan amount, repayment history, employment status, and loan approval status. The bank wants to understand customer risk and improve loan approval decisions.

Questions:

- A. How can descriptive statistics be used to summarize customer income, credit scores, and loan amounts?
- B. Discuss how unusually high loan amounts or very low credit scores can influence the analysis. Explain methods for identifying such outliers.
- C. How can hypothesis testing be used to determine whether customers with higher credit scores are more likely to receive loan approval than customers with lower credit scores? Formulate the hypotheses and describe the analysis process.

Answer:

A. Descriptive Statistics

Descriptive statistics help summarize customer financial data. The **mean** shows the average

income, credit score, and loan amount, the **median** provides the middle value, **standard deviation** measures the variation in these values, and **quartiles** divide the data into four parts to understand the distribution and identify customer risk levels.

B. Impact of Outliers and Detection Methods

Unusually high loan amounts or very low credit scores can distort the analysis and affect loan approval decisions. These outliers may lead to inaccurate estimates of customer risk. Outliers can be detected using the **Interquartile Range (IQR) method** or the **Z-score method**. After detection, they can be verified, corrected if due to data errors, or analyzed separately for better decision-making.

C. Hypothesis Testing

Hypothesis testing is used to determine whether customers with higher credit scores are more likely to receive loan approval.

- **Null Hypothesis (H_0):** Customers with higher credit scores are not more likely to receive loan approval.
- **Alternative Hypothesis (H_1):** Customers with higher credit scores are more likely to receive loan approval.

The data is divided into customers with **higher** and **lower** credit scores, and an appropriate statistical test (such as an **Independent Sample t-test**) is performed. The **p-value** is compared with **0.05**. If $p < 0.05$, reject H_0 and conclude that higher credit scores significantly increase the likelihood of loan approval. Otherwise, fail to reject H_0 .

Case Study 6: Online Learning Platform Analysis

Scenario:

An online learning platform has collected learner activity data, including course completion percentage, quiz scores, video watch time, login frequency, and feedback ratings. The platform wants to improve course engagement and learner success.

Questions:

- A. Explain how descriptive statistics can be used to summarize quiz scores, watch time, and course completion rates.
- B. Discuss how unusually high or low learning activity values may affect the analysis. What techniques can be used to detect outliers?
- C. How can hypothesis testing be used to determine whether learners who watch more than 70% of the course videos score significantly higher in quizzes than other learners? Formulate the hypotheses and explain the testing procedure.

A. Descriptive Statistics

Descriptive statistics help summarize learner activity data. The **mean** shows the average quiz scores, watch time, and course completion rate, while the **median** gives the middle value. **Standard deviation** measures the variation in learner performance, and **quartiles** divide the data into four parts to identify different learner groups and overall engagement.

B. Impact of Outliers and Detection Methods

Unusually high or low learning activity values can distort the average and lead to incorrect conclusions about learner engagement. Outliers can be detected using the **Interquartile Range (IQR) method** or the **Z-score method**. After detection, they should be verified and corrected, removed, or analyzed separately if necessary.

C. Hypothesis Testing

Hypothesis testing is used to determine whether learners who watch more than **70%** of the course videos score significantly higher in quizzes.

- **Null Hypothesis (H_0):** There is no significant difference in quiz scores between learners who watch more than 70% of videos and others.
- **Alternative Hypothesis (H_1):** Learners who watch more than 70% of videos score significantly higher in quizzes.

An **Independent Sample t-test** is performed, and the **p-value** is compared with **0.05**. If $p < 0.05$, reject H_0 ; otherwise, fail to reject H_0 .

Case Study 7: Retail Sales Performance Analysis

Scenario:

A retail chain has collected sales data from different branches, including daily sales amount, number of customers, discount percentage, product category, and customer feedback score.

The company wants to evaluate branch performance and improve sales planning. **Questions:**

A. How can descriptive statistics and distribution analysis be used to summarize daily sales and customer count?

B. Discuss how abnormal sales values, such as festival-season sales or very low sales days, can affect the analysis. Explain suitable outlier detection methods.

C. How can confidence intervals and hypothesis testing be used to determine whether the average sales of urban branches are significantly higher than rural branches?

A. Descriptive Statistics and Distribution Analysis

Descriptive statistics summarize daily sales and customer count. The **mean** gives the average sales and customer visits, the **median** shows the middle value, **standard deviation** measures

variation, and **quartiles** divide the data into four parts. **Distribution analysis** helps identify whether sales are normally distributed or influenced by seasonal trends.

B. Impact of Outliers and Detection Methods

Abnormal sales during festivals or very low sales days can affect the average sales and increase data variability. These outliers can be detected using the **IQR method** or **Z-score method**. After detection, they should be verified and analyzed separately or removed if caused by errors.

C. Confidence Intervals and Hypothesis Testing

A **confidence interval** estimates the range in which the true average sales are likely to lie.

Hypothesis testing compares the average sales of urban and rural branches.

- **Null Hypothesis (H_0):** Urban and rural branches have the same average sales.
- **Alternative Hypothesis (H_1):** Urban branches have significantly higher average sales than rural branches.

An **Independent Sample t-test** is performed. If the **p-value** < **0.05**, reject H_0 ; otherwise, fail to reject H_0 .

Case Study 8: Food Delivery Service Analysis

Scenario:

A food delivery company has collected order data, including delivery time, order value, customer rating, distance travelled, and delivery partner experience. The company wants to reduce delivery delays and improve customer satisfaction.

Questions:

A. Explain how descriptive statistics such as mean, median, standard deviation, and quartiles can be used to analyze delivery time and customer ratings.

B. Discuss how unusually long delivery times can affect customer satisfaction analysis.

Describe suitable methods for detecting outliers.

C. How can hypothesis testing be used to determine whether experienced delivery partners complete deliveries faster than new delivery partners? Formulate the hypotheses and explain the analysis process.

A. Descriptive Statistics

Descriptive statistics summarize delivery performance. The **mean** gives the average delivery time and customer rating, the **median** shows the middle value, **standard deviation** measures the variation, and **quartiles** help identify the spread of delivery times and customer satisfaction.

B. Impact of Outliers and Detection Methods

Unusually long delivery times can reduce the average customer satisfaction and affect performance analysis. Outliers can be detected using the **Interquartile Range (IQR)** or **Z-score** method. Once identified, they should be verified and handled appropriately to improve analysis accuracy.

C. Hypothesis Testing

Hypothesis testing is used to determine whether experienced delivery partners complete deliveries faster than new delivery partners.

- **Null Hypothesis (H_0):** There is no significant difference in delivery time between experienced and new delivery partners.
- **Alternative Hypothesis (H_1):** Experienced delivery partners complete deliveries significantly faster.

An **Independent Sample t-test** is conducted, and the **p-value** is compared with **0.05**. If **p < 0.05**, reject H_0 ; otherwise, fail to reject H_0 .

Case Study 9: Manufacturing Defect Analysis

Scenario:

A manufacturing company has collected production data, including number of units produced, defect count, machine temperature, production time, and operator shift. The company wants to reduce defects and improve product quality.

Questions:

- A. How can descriptive statistics be used to summarize defect counts, production time, and machine temperature?
- B. Discuss how abnormal defect values may affect the quality analysis. What methods can be used to detect outliers in production data?
- C. How can hypothesis testing be used to determine whether the defect rate differs significantly between day shift and night shift production? Formulate the hypotheses and explain the analysis process.

A. Descriptive Statistics

Descriptive statistics summarize production data. The **mean** shows the average defect count, production time, and machine temperature. The **median** provides the middle value, **standard deviation** measures variation, and **quartiles** help understand the distribution and identify unusual production values.

B. Impact of Outliers and Detection Methods

Abnormal defect values can distort quality analysis and lead to incorrect conclusions about

production performance. Outliers can be detected using the **IQR** and **Z-score** methods. After detection, they should be verified and corrected or analyzed separately if caused by special production conditions.

C. Hypothesis Testing

Hypothesis testing is used to determine whether the defect rate differs between day and night shifts.

- **Null Hypothesis (H_0):** There is no significant difference in defect rate between day and night shifts.
- **Alternative Hypothesis (H_1):** The defect rate differs significantly between day and night shifts.

An **Independent Sample t-test** is performed, and the **p-value** is compared with **0.05**. If $p < 0.05$, reject H_0 ; otherwise, fail to reject H_0 .

Case Study 10: Mobile Network Performance Analysis

Scenario:

A telecom company has collected mobile network data, including signal strength, internet speed, call drop rate, number of users, and customer complaints from different locations. The company wants to improve network quality.

Questions:

- Explain how descriptive statistics and distribution analysis can be used to summarize internet speed, signal strength, and call drop rate.
- Discuss how extremely low internet speed or high call drop rate values can affect the analysis. Describe suitable outlier detection and handling methods.
- How can confidence intervals and hypothesis testing be used to determine whether the average internet speed in urban areas is significantly different from rural areas?

Answer :

A. Descriptive Statistics and Distribution Analysis

Descriptive statistics summarize internet speed, signal strength, and call drop rate. The **mean** gives the average performance, the **median** provides the middle value, **standard deviation** measures variability, and **quartiles** show the spread of the data. **Distribution analysis** helps identify performance trends and unusual network behavior.

B. Impact of Outliers and Detection Methods

Extremely low internet speeds or high call drop rates can affect the overall network analysis

and lead to misleading results. Outliers can be detected using the **Interquartile Range (IQR)** and **Z-score** methods. They should be verified and corrected, removed, or analyzed separately based on the cause.

C. Confidence Intervals and Hypothesis Testing

A **confidence interval** estimates the range in which the true average internet speed is expected to lie. Hypothesis testing compares the average internet speed between urban and rural areas.

- **Null Hypothesis (H_0):** There is no significant difference in average internet speed between urban and rural areas.
- **Alternative Hypothesis (H_1):** The average internet speed in urban areas is significantly different from that in rural areas.

An **Independent Sample t-test** is conducted, and the **p-value** is compared with **0.05**. If $p < 0.05$, reject H_0 ; otherwise, fail to reject H_0 .

Case Study 11: Public Transportation Usage Analysis

Scenario:

A city transport department has collected bus usage data, including passenger count, ticket revenue, waiting time, travel duration, and route distance. The department wants to improve transport scheduling and passenger service.

Questions:

- A. How can descriptive statistics be used to summarize passenger count, waiting time, and travel duration?
- B. Discuss how unusually high passenger counts during peak hours or unusually long waiting times can affect the analysis. Explain methods for detecting outliers.
- C. How can hypothesis testing be used to determine whether average waiting time during peak hours is significantly higher than during non-peak hours? Formulate the hypotheses and explain the testing process.

Answer :

A. Descriptive Statistics

Descriptive statistics help summarize passenger count, waiting time, and travel duration. The **mean** shows the average values, the **median** gives the middle value, **standard deviation** measures the variation, and **quartiles** divide the data into four parts to understand the distribution and identify unusual travel patterns.

B. Impact of Outliers and Detection Methods

Unusually high passenger counts during peak hours or very long waiting times can distort the average and affect transport planning. Outliers can be detected using the **Interquartile Range (IQR) method** or the **Z-score method**. After detection, they should be verified and analyzed separately or corrected if caused by data errors.

C. Hypothesis Testing

Hypothesis testing is used to determine whether the average waiting time during peak hours is significantly higher than during non-peak hours.

- **Null Hypothesis (H_0):** There is no significant difference in average waiting time between peak and non-peak hours.
- **Alternative Hypothesis (H_1):** The average waiting time during peak hours is significantly higher.

An **Independent Sample t-test** is performed, and the **p-value** is compared with **0.05**. If **p < 0.05**, reject H_0 ; otherwise, fail to reject H_0 .

Case Study 12: Agriculture Crop Yield Analysis

Scenario:

An agriculture department has collected crop production data from different farms, including rainfall, fertilizer usage, soil moisture, temperature, and crop yield. The department wants to analyze factors affecting crop productivity.

Questions:

- Explain how descriptive statistics and distribution analysis can be used to summarize crop yield, rainfall, and fertilizer usage.
- Discuss how abnormal crop yield values caused by drought, pests, or excessive rainfall can affect the analysis. Describe suitable outlier detection methods.
- How can confidence intervals and hypothesis testing be used to determine whether farms using organic fertilizer have significantly different crop yield compared with farms using chemical fertilizer?

A. Descriptive Statistics and Distribution Analysis

Descriptive statistics summarize crop yield, rainfall, and fertilizer usage. The **mean** gives the average values, the **median** provides the middle value, **standard deviation** measures variation, and **quartiles** show the spread of the data. **Distribution analysis** helps identify patterns, skewness, and unusual crop yield values across farms.

B. Impact of Outliers and Detection Methods

Abnormal crop yields caused by drought, pests, or excessive rainfall can affect the overall analysis and produce misleading results. Outliers can be detected using the **IQR** and **Z-score**

methods. Once identified, they should be verified and either analyzed separately or removed if they result from data errors.

C. Confidence Intervals and Hypothesis Testing

A **confidence interval** estimates the range within which the true average crop yield is expected to lie. Hypothesis testing compares the crop yields of farms using organic fertilizer and chemical fertilizer.

- **Null Hypothesis (H_0):** There is no significant difference in crop yield between organic and chemical fertilizers.
- **Alternative Hypothesis (H_1):** Crop yield differs significantly between organic and chemical fertilizers.

An **Independent Sample t-test** is conducted. If the **p-value** < 0.05 , reject H_0 ; otherwise, fail to reject H_0 .

Case Study 13: Employee Productivity Analysis

Scenario:

A software company has collected employee productivity data, including working hours, number of completed tasks, project deadlines, experience level, and performance rating. The company wants to improve employee performance and project planning. **Questions:**

A. How can descriptive statistics such as mean, median, standard deviation, and quartiles be used to summarize employee productivity data?

B. Discuss how unusually high or low task completion values can affect productivity analysis. Describe suitable methods for detecting outliers.

C. How can hypothesis testing be used to determine whether experienced employees complete significantly more tasks than newly joined employees? Formulate the hypotheses and explain the analysis process.

A. Descriptive Statistics

Descriptive statistics help summarize employee productivity data. The **mean** shows the average number of completed tasks and working hours, the **median** gives the middle value, **standard deviation** measures the variation in productivity, and **quartiles** divide the data into four parts to identify different performance levels.

B. Impact of Outliers and Detection Methods

Unusually high or low task completion values can distort productivity analysis and lead to incorrect performance evaluation. Outliers can be detected using the **Interquartile Range (IQR) method** or the **Z-score method**. After detection, they should be verified and corrected,

removed, or analyzed separately.

C. Hypothesis Testing

Hypothesis testing is used to determine whether experienced employees complete significantly more tasks than newly joined employees.

- **Null Hypothesis (H_0):** There is no significant difference in the number of completed tasks between experienced and newly joined employees.
- **Alternative Hypothesis (H_1):** Experienced employees complete significantly more tasks than newly joined employees.

An **Independent Sample t-test** is performed, and the **p-value** is compared with **0.05**. If **p < 0.05**, reject H_0 and conclude that experienced employees complete significantly more tasks. Otherwise, fail to reject H_0 .

Case Study 14: Automobile Fuel Efficiency Analysis

Scenario:

An automobile company has collected vehicle performance data, including fuel consumption, engine capacity, vehicle weight, mileage, and maintenance cost. The company wants to analyze fuel efficiency and improve vehicle design.

Questions:

- Explain how descriptive statistics and distribution analysis can be used to summarize mileage and fuel consumption data.
- Discuss how abnormal mileage values can affect the analysis. What methods can be used to detect and handle outliers?
- How can confidence intervals and hypothesis testing be used to determine whether hybrid cars have significantly better mileage than petrol cars?

A. Descriptive Statistics and Distribution Analysis

Descriptive statistics help summarize mileage and fuel consumption data. The **mean** shows the average mileage and fuel usage, the **median** provides the middle value, **standard deviation** measures the variation, and **quartiles** divide the data into four parts to understand its spread. **Distribution analysis** helps identify patterns, skewness, and overall vehicle performance.

B. Impact of Outliers and Detection Methods

Abnormal mileage values can distort the average fuel efficiency and lead to incorrect conclusions about vehicle performance. Outliers can be detected using the **Interquartile Range (IQR) method** or the **Z-score method**. After detection, they should be verified and

corrected, removed, or analyzed separately if they are genuine extreme cases.

C. Confidence Intervals and Hypothesis Testing

A **confidence interval** estimates the range in which the true average mileage is expected to lie. Hypothesis testing compares the mileage of hybrid and petrol cars.

- **Null Hypothesis (H_0):** There is no significant difference in mileage between hybrid and petrol cars.
- **Alternative Hypothesis (H_1):** Hybrid cars have significantly better mileage than petrol cars.

An **Independent Sample t-test** is performed, and the **p-value** is compared with **0.05**. If **p < 0.05**, reject H_0 ; otherwise, fail to reject H_0 .

Case Study 15: Social Media Engagement Analysis

Scenario:

A digital marketing agency has collected social media data, including number of likes, shares, comments, followers, posting time, and engagement rate. The agency wants to improve content strategy and audience reach.

Questions:

- A. How can descriptive statistics be used to summarize likes, shares, comments, and engagement rate?
- B. Discuss how viral posts with extremely high engagement values may affect the analysis. Describe suitable outlier detection methods.
- C. How can hypothesis testing be used to determine whether video posts receive significantly higher engagement than image posts? Formulate the hypotheses and explain the testing procedure.

A. Descriptive Statistics

Descriptive statistics help summarize social media performance. The **mean** shows the average number of likes, shares, comments, and engagement rate, the **median** provides the middle value, **standard deviation** measures the variation in engagement, and **quartiles** divide the data into four parts to identify highly and less engaging posts.

B. Impact of Outliers and Detection Methods

Viral posts with extremely high engagement can increase the average and make normal posts appear less effective. Outliers can be detected using the **Interquartile Range (IQR) method** or the **Z-score method**. Once detected, they should be verified and analyzed separately or handled appropriately to improve the accuracy of the analysis.

C. Hypothesis Testing

Hypothesis testing is used to determine whether video posts receive significantly higher engagement than image posts.

- **Null Hypothesis (H_0):** There is no significant difference in engagement between video posts and image posts.
- **Alternative Hypothesis (H_1):** Video posts receive significantly higher engagement than image posts.

An **Independent Sample t-test** is conducted, and the **p-value** is compared with **0.05**. If **p < 0.05**, reject H_0 and conclude that video posts receive significantly higher engagement. Otherwise, fail to reject H_0 .

Case Study 16: Airline Passenger Satisfaction Analysis

Scenario:

An airline company has collected passenger feedback data, including ticket price, flight delay time, baggage waiting time, seat comfort rating, and overall satisfaction score. The airline wants to improve service quality.

Questions:

- Explain how descriptive statistics can be used to summarize delay time, baggage waiting time, and passenger satisfaction scores.
- Discuss how unusually long flight delays can affect the analysis of passenger satisfaction. What methods can be used to detect outliers?
- How can hypothesis testing be used to determine whether passengers on on-time flights give significantly higher satisfaction ratings than passengers on delayed flights?

A. Descriptive Statistics

Descriptive statistics help summarize delay time, baggage waiting time, and passenger satisfaction scores. The **mean** shows the average values, the **median** provides the middle value, **standard deviation** measures the variation, and **quartiles** divide the data into four parts to understand the distribution and identify unusual service patterns.

B. Impact of Outliers and Detection Methods

Unusually long flight delays can reduce the average passenger satisfaction and distort the overall analysis. Outliers can be detected using the **Interquartile Range (IQR) method** or the **Z-score method**. Once identified, they should be verified and either corrected, removed if caused by data errors, or analyzed separately if they represent actual events.

C. Hypothesis Testing

Hypothesis testing is used to determine whether passengers on on-time flights give significantly higher satisfaction ratings than passengers on delayed flights.

- **Null Hypothesis (H_0):** There is no significant difference in satisfaction ratings between passengers on on-time flights and delayed flights.
- **Alternative Hypothesis (H_1):** Passengers on on-time flights give significantly higher satisfaction ratings than passengers on delayed flights.

An **Independent Sample t-test** is performed by comparing the satisfaction scores of the two groups. The **p-value** is compared with **0.05**. If $p < 0.05$, reject H_0 and conclude that on-time flights result in significantly higher passenger satisfaction. Otherwise, fail to reject H_0 .

Case Study 17: Smart City Traffic Analysis

Scenario:

A smart city traffic department has collected traffic data, including vehicle count, average speed, signal waiting time, accident count, and pollution level from different roads. The department wants to reduce congestion and improve road safety.

Questions:

- A. How can descriptive statistics and distribution analysis be used to summarize vehicle count, waiting time, and average speed?
- B. Discuss how abnormal traffic values during accidents, festivals, or roadblocks can affect the analysis. Describe suitable methods for detecting outliers.
- C. How can hypothesis testing be used to determine whether average waiting time is significantly higher during peak hours than during non-peak hours?

A. Descriptive Statistics and Distribution Analysis

Descriptive statistics help summarize vehicle count, waiting time, and average speed. The **mean** shows the average traffic flow and waiting time, the **median** provides the middle value, **standard deviation** measures the variation in traffic conditions, and **quartiles** divide the data into four parts to understand its spread. **Distribution analysis** helps identify traffic patterns, congestion levels, and unusual road conditions.

B. Impact of Outliers and Detection Methods

Abnormal traffic values caused by accidents, festivals, or roadblocks can distort the average traffic flow and waiting time, leading to inaccurate traffic analysis. Outliers can be detected using the **Interquartile Range (IQR) method** or the **Z-score method**. After detection, they should be verified and analyzed separately or removed if they are due to data errors.

C. Hypothesis Testing

Hypothesis testing is used to determine whether the average waiting time during peak hours is significantly higher than during non-peak hours.

- **Null Hypothesis (H_0):** There is no significant difference in average waiting time between peak and non-peak hours.
- **Alternative Hypothesis (H_1):** The average waiting time during peak hours is significantly higher than during non-peak hours.

An **Independent Sample t-test** is performed by comparing the waiting times of the two groups. The **p-value** is compared with **0.05**. If $p < 0.05$, reject H_0 and conclude that waiting time is significantly higher during peak hours. Otherwise, fail to reject H_0 .

Case Study 18: Hotel Booking and Customer Review Analysis

Scenario:

A hotel chain has collected booking data, including room price, length of stay, customer rating, booking cancellation status, and service feedback score. The hotel wants to improve customer satisfaction and pricing strategy.

Questions:

- A. How can descriptive statistics be used to summarize room price, length of stay, and customer ratings?
- B. Discuss how unusually high room prices or extremely low customer ratings can affect the analysis. Explain suitable methods for detecting outliers.
- C. How can hypothesis testing be used to determine whether customers who stay more than three days give significantly different ratings compared with customers who stay for one or two days?

A. Descriptive Statistics

Descriptive statistics help summarize room price, length of stay, and customer ratings. The **mean** shows the average room price and stay duration, the **median** provides the middle value, **standard deviation** measures the variation in prices and ratings, and **quartiles** divide the data into four parts to understand the distribution and identify different customer groups.

B. Impact of Outliers and Detection Methods

Unusually high room prices or extremely low customer ratings can distort the average and affect the analysis of customer satisfaction and pricing strategy. Outliers can be detected using the **Interquartile Range (IQR) method** or the **Z-score method**. After detection, they should be verified and either corrected, removed if caused by errors, or analyzed separately if they represent genuine cases.

C. Hypothesis Testing

Hypothesis testing is used to determine whether customers who stay for more than **three days** give significantly different ratings compared with customers who stay for **one or two days**.

- **Null Hypothesis (H_0):** There is no significant difference in customer ratings between the two groups.
- **Alternative Hypothesis (H_1):** Customers who stay for more than three days give significantly different ratings.

An **Independent Sample t-test** is performed by comparing the ratings of both groups. The **p-value** is compared with **0.05**. If $p < 0.05$, reject H_0 and conclude that the length of stay significantly affects customer ratings. Otherwise, fail to reject H_0 .

Case Study 19: Fitness App User Health Analysis

Scenario:

A fitness app has collected user health data, including daily steps, calories burned, workout duration, heart rate, sleep hours, and fitness goal completion. The app wants to study user activity patterns and improve health recommendations.

Questions:

- A. Explain how descriptive statistics can be used to summarize daily steps, calories burned, workout duration, and sleep hours.
- B. Discuss how unusually high workout duration or abnormal heart rate values can affect the health analysis. What methods can be used to detect outliers?
- C. How can hypothesis testing be used to determine whether users who sleep at least seven hours per day burn significantly more calories than users who sleep less than seven hours?

A. Descriptive Statistics

Descriptive statistics help summarize daily steps, calories burned, workout duration, and sleep hours. The **mean** shows the average values, the **median** provides the middle value, **standard deviation** measures the variation in user activity, and **quartiles** divide the data into four parts to understand activity levels and overall health patterns.

B. Impact of Outliers and Detection Methods

Unusually high workout duration or abnormal heart rate values can distort the average and lead to incorrect health recommendations. Outliers can be detected using the **Interquartile Range (IQR) method** or the **Z-score method**. After detection, they should be verified and either corrected, removed if caused by errors, or analyzed separately if they represent genuine health conditions.

C. Hypothesis Testing

Hypothesis testing is used to determine whether users who sleep **at least seven hours** burn significantly more calories than users who sleep **less than seven hours**.

- **Null Hypothesis (H_0):** There is no significant difference in calories burned between the two groups.
- **Alternative Hypothesis (H_1):** Users who sleep at least seven hours burn significantly more calories than users who sleep less than seven hours.

An **Independent Sample t-test** is performed, and the **p-value** is compared with **0.05**. If **p < 0.05**, reject H_0 and conclude that adequate sleep significantly affects calories burned.

Otherwise, fail to reject H_0 .

Case Study 20: Online Examination Performance Analysis

Scenario:

An educational institution has collected online exam data, including exam score, time taken to complete the test, number of attempts, internet interruption count, and student feedback. The institution wants to evaluate student performance and improve the online assessment system.

Questions:

- A. How can descriptive statistics and distribution analysis be used to summarize exam scores, completion time, and number of attempts?
- B. Discuss how unusually short completion time or extremely high number of attempts can affect exam performance analysis. Describe suitable outlier detection methods.
- C. How can hypothesis testing be used to determine whether students with fewer internet interruptions score significantly higher than students with frequent interruptions?

A. Descriptive Statistics and Distribution Analysis

Descriptive statistics help summarize exam scores, completion time, and number of attempts.

The **mean** shows the average values, the **median** provides the middle value, **standard deviation** measures the variation, and **quartiles** divide the data into four parts to understand the distribution. **Distribution analysis** helps identify performance patterns, skewness, and unusual exam behavior.

B. Impact of Outliers and Detection Methods

Unusually short completion times or an extremely high number of attempts can distort the analysis and lead to incorrect conclusions about student performance. Outliers can be detected using the **Interquartile Range (IQR) method** or the **Z-score method**. Once identified, they should be verified and handled by correcting errors, removing invalid values, or analyzing them separately.

C. Hypothesis Testing

Hypothesis testing is used to determine whether students with **fewer internet interruptions** score significantly higher than students with **frequent interruptions**.

- **Null Hypothesis (H_0):** There is no significant difference in exam scores between students with fewer and frequent internet interruptions.
- **Alternative Hypothesis (H_1):** Students with fewer internet interruptions score significantly higher.

An **Independent Sample t-test** is performed by comparing the exam scores of the two groups. The **p-value** is compared with **0.05**. If $p < 0.05$, reject H_0 and conclude that fewer internet interruptions lead to significantly higher exam scores. Otherwise, fail to reject H_0 .